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Studierichting: Informatica
Oefeningenreeks 4A nr. 30.5

$$\int 3x\sqrt{4x^2 - 1}dx$$
$$= 3 \int x\sqrt{4x^2 - 1}dx$$

$$\text{Stel } u = 4x^2 - 1 \Rightarrow du = 8xdx \Rightarrow \frac{1}{8}du = xdx$$

$$= 3 \cdot \frac{1}{8} \int \sqrt{u}du$$

$$= \frac{3}{8} \cdot \frac{2}{3} \cdot u^{\frac{3}{2}} + C$$

$$= \frac{1}{4}(4x^2 - 1)^{\frac{3}{2}} + C$$

References